Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/14/2005
PI Name:	Drysdale, Alan E Ph.D.		
Project Title:	Equivalent System Mass Impacts of Clothing on Wastes and Mission Cost		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32815	**Congressional District:**	24
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	03/03/2003	**End Date:**	03/02/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	4
No. of PhD Candidates:	4	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Garton, Harry (Boeing) Hasselbrack, Sally (Boeing) Barker, Roger (NCSU) Thompson, Don (NCSU) Scruggs, Barbara (NCSU) Fratto, Lisa (NCSU)		
Grant/Contract No.:			
Performance Goal No.:			
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Task Description:	Resupply of clothing comprises about 16% - 19% of the mission ESM for long duration missions. Disposing of the clothing as waste also constitutes a significant impact. Two approaches have been identified that show promise for large reductions in these impacts. · Mass of clothes can be reduced by using high-tech materials, which also provide other comfort and safety benefits. · Mass of clothes can be reduced by using a cleaning system and reusing clothes. This does require additional equipment and interfaces, but preliminary analyses show promise that the net ESM would be greatly reduced. For the International Space Station (ISS), crew clothing is supplied at a rate of about 194 kg and 1 m3 per 90-day increment for a crew of 3, though an attempt to reduce this is underway. The clothing, once worn, then becomes waste that must be managed. Crew debriefs (Increment 1) documented that odors produced from soiled clothing were more unpleasant for the crew than were odors from food trash. We propose to evaluate standards and requirements for providing clothing for human spaceflight missions, using ISS as a baseline and modifying and expanding the requirements for future missions. We will address two possible ways to reduce the mission impact of clothing; a redesign of the wardrobe, and cleaning and re-use of clothing in space. We will identify life-support related impacts that clothing has on subsystems and interfaces, options for processing the clothing, costs and benefits, and recommend particular approaches for particular near-term missions. Reduce clothing system ESM. The clothing system is one of the few areas where significant improvements in the ALS project metric can be made.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Potential cross utilization of developed technology. In particular, reduced water and energy use, and lower vibration and noise, are likely.
Task Progress:	Considerable work was done on laboratory testing of tee shirt fabrics at NCSU to determine if better choices are available than the cotton currently in use. Fabrics were found that did score better overall, and in specific areas such as flammability. Ranking of the different fabrics tested varied with the weightings, though only one fabric changed ranking dramatically. This was a very comfortable but bulky and heavy fabric. Estimates of clothing system ESM were refined. Fabric selection could reduce ESM by about a factor of 2. Clothes washing could reduce ESM by more than an order of magnitude for a long duration mission. A clothes dryer would be cost effective on a missionlasting longer than a few weeks. A clothes washer would be cost effective for a mission that was not much longer. Cleaning clothes by other methods, such as using liquid CO2 was judged to be not competitive, due to system mass and concerns over accidental leakage into a closed habitat. Breakeven for cleaning clothing depends heavily on the cost of water regeneration, and the trade study must be done at a system level. It also depends on how much there is to be washed. Besides IVA clothing, some EVA clothing (liquid cooled garments and comfort undergarments), towels, bedding, and some wipes could also profitably be cleaned. Initial assessments of clothing washing and drying in different gravity environments were made. It was considered probable that a COTS design could be modified to provide a cost effective approach to clothes cleaning in the gravity environments expected. Outreach included working with the Purdue NSCORT and other university teams. This work was coordinated with the SIMA team at NASA JSC.
Bibliography Type:	Description: (Last Updated: 04/14/2021)
Papers from Meeting Proceedings	Drysdale AE. "Equivalencies and Applicability of Advanced Life Support Technologies to Exploration Missions." International Conference on Environmental Systems, Rome, July 2005. International Conference on Environmental Systems, Rome, July 2005. SAE paper 2005-01-2915. https://doi.org/10.4271/2005-01-2915 , Jul-2005